Reliable Annotation and Evaluation of Malaysian Malay-English Code-Mixed Sentiment Data: A Critical Review of an Unresolved Problem

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Abstract

Malaysian social media users mix English and Malay inside single messages, producing the contact varieties known as Bahasa Rojak and Manglish. The studies reviewed here usually treat sentiment analysis of Malay-English mixed writing as a modelling problem. This review indicates that the more limiting problem originates earlier, in the sentiment labels themselves and in the evidence offered that those labels are sound. The review covers annotation practice and reliability evaluation for Malaysian Malay-English code-mixed sentiment data. Comparative benchmarking of sentiment classifiers is outside its scope. Three contributions follow. First, the annotation strategies used for low-resource code-mixed text are organised into three tiers, and each tier is shown to improve either label fidelity or scalability at the expense of the other. Second, the two causes of neutral-label skew are separated, one produced by the labelling tools and the other by genuine uncertainty among annotators. The four regional corpora are then examined against what a reliability claim requires, and none supplies it. Third, the class imbalance in the label distribution is shown to be the same condition under which Cohen's kappa becomes difficult to interpret. The review closes with a research agenda covering minimum reporting standards, agreement measurement under class imbalance, and the untested assumption that annotators of different types make different errors.

Keywords: *sentiment analysis, code-mixed text, Bahasa Rojak, annotation reliability, inter-annotator agreement*

1. Introduction

Sentiment analysis, also known as opinion mining, is the automated task of deciding whether a piece of text expresses a positive, negative or neutral opinion (Poria et al., 2023). Organisations use it to measure public opinion at scale, in applications from brand monitoring to political and policy analysis (Poria et al., 2023; Nazir et al., 2022). The tools that make this possible carry an assumption about their input: lexicons, pre- trained models and benchmark corpora are concentrated in a small number of well-resourced languages, each built from text written in a single standard variety (Joshi et al., 2020). Text in which two languages alternate within one message does not satisfy that assumption, and surveys of code-switching in natural language processing report it as a persistent source of degraded performance (Doğruöz et al., 2021; Winata et al., 2023). Malaysian social media text is writing of exactly this kind.

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Malaysia is a multilingual and multi-ethnic country in which Coluzzi (2017) recorded around 140 languages in use, with code-mixing between a speaker's first language, English and Malay already widespread in everyday communication. The practice has continued rather than receded, and on social media users alternate between languages within a single message rather than writing in one language at a time (Treffers-Daller et al., 2022; AlAfnan et al., 2025). English and Malay are the pair that combines most frequently in Malaysian writing, and the mixed varieties they produce are known locally as *Bahasa Rojak* and *Manglish* (AlAfnan et al., 2025). Malay-English mixed writing is frequent enough on X, the platform formerly known as Twitter, that both of the Malaysian sentiment corpora examined in this review were collected from it (Maskat et al., 2024; Mohd Shamsuddin et al., 2024).

How this mixed writing is regarded matters for how it is processed. *Manglish* is sometimes described as a variety that weakens standard English and Malay rather than as a legitimate language variety in its own right (AlAfnan et al., 2025). Its structure, however, is not random. Treffers-Daller et al. (2022) and AlAfnan et al. (2025) show that Malay-English switching follows describable syntactic and pragmatic patterns, and that speakers use it deliberately to communicate efficiently, signal identity and build rapport. The same regularity holds at the level of spelling: Maskat and Rahman (2020) record the negator *tidak* written as *x*, and *tidak ada* as *takde*, *xde* or *tade*. A reader who knows the convention recovers the word without effort, but a lexicon compiled from standard Malay lists none of those forms. The difficulty that Malay- English mixed writing creates for natural language processing (NLP) is therefore not carelessness. It is that the conventions Malaysian writers follow are not the conventions standard NLP tools were built to handle, so writing that is systematic is processed as noise (Winata et al., 2023; Abu Bakar et al., 2020).

When a sentiment model performs poorly on Malay-English mixed text, the explanation offered is most often architectural (Doğruöz et al., 2021; Winata et al., 2023). On that account the model cannot handle code- switched input, and a larger or more regionally adapted model would perform better. The evidence is mixed. Khanuja et al. (2020) report that multilingual models fine-tuned on code-switched data achieve the strongest scores among the systems evaluated on their benchmark. Adilazuarda et al. (2022) show that robustness depends on how well each language is represented during pre-training. Against that, Winata et al. (2021) find that multilingual pre-training alone does not guarantee good representations of code-switched text. Conneau et al. (2020) describe a "curse of multilinguality" in which finite model capacity, spread across many languages, dilutes per-language quality. Model choice therefore accounts for part of the difficulty but not all of it.

The remainder of the difficulty does not originate in the model. It originates one step earlier, at the point where the labels are created. Supervised sentiment analysis requires that every message be read and assigned a sentiment label, by a human annotator or by an automatic one, before any model can be trained or evaluated on it (Artstein and Poesio, 2008). This step is called annotation, and the labels it produces form the supervision signal that a model learns from and is measured against. The consequences of incorrect labels reach beyond the dataset. Label noise is learned during training and degrades the resulting classifier (Frénay and Verleysen, 2014). Incorrect labels in a test set also distort the evaluation of model

performance, so that reported accuracy no longer reflects how the model actually behaves (Northcutt et al., 2021; Bowman and Dahl, 2021). An unsound label therefore damages the model twice, once when it is trained and again when it is measured.

This review indicates that for Malaysian Malay-English code-mixed sentiment data, the annotation label is at present the more limiting problem, together with the evidence offered that the label is sound. Three findings support that reading, and each is developed in the sections that follow. First, none of the four existing Malay-English sentiment resources is simultaneously large, balanced across sentiment classes, drawn from casual social media and validated by human annotators; each secures some of these properties at the expense of the others. Second, the class imbalance in the largest of them originates in the labelling method rather than in Malaysian discourse. Third, annotation reliability is rarely reported, and where a coefficient is given it was obtained for a narrower task than the corpus is used to support.

The second and third findings describe one condition rather than two. The skew the labelling methods produce is itself the condition under which a chance-corrected coefficient becomes difficult to interpret, so correcting either weakness alone leaves the other in place. A 2024 review of the field still reports a lack of comprehensive datasets and annotated resources for code-mixed Malay text (Selvaraju et al., 2024).

This review examines one question: why reliable annotation and evaluation of Malaysian Malay-English code- mixed sentiment data remain unresolved. Linguistic description, model architecture and application domain are discussed only where they are relevant to that question. Two related topics fall outside the scope. The first is comparative benchmarking of sentiment classifiers, and the second is sentiment analysis as a general field. Both have already been surveyed in detail (Poria et al., 2023; Nazir et al., 2022; Selvaraju et al., 2024). This review does not propose, implement or evaluate a system, and it does not report the author's own experimental work. Its contribution is a diagnosis of why the problem remains unresolved, together with a set of directions for future research.

This review makes three contributions:

1. It organises the annotation strategies used for low-resource code-mixed text into three tiers, which are manual, semi-automated and fully automated labelling, and shows that each tier gains either label fidelity or scalability at the expense of the other.
2. It distinguishes the two causes of neutral-label skew, one produced by the labelling tools themselves and the other by genuine uncertainty among annotators, and it examines what each of the four regional corpora reports about its own reliability.
3. It shows how the linguistic, annotation and measurement challenges compound one another rather than merely co-occurring, and sets out a research agenda that follows from that interaction.

The literature examined here was identified through searches of the ACL Anthology, IEEE Xplore, Scopus and arXiv, combining terms for the language

variety, the sentiment task and annotation quality. Studies that only benchmark classifiers, with no annotation or reliability component, were not included.

The rest of this paper is organised as follows. Section 2 explains why Malaysian code-mixed writing is difficult to annotate. Section 3 examines annotation practice and its failure modes, and Section 4 examines the models that are used as annotators. Section 5 discusses how annotation reliability is evaluated. Section 6 brings these findings together and sets out directions for future research, and Section 7 concludes the review.

2. Why Malaysian Code-Mixed Text Is Difficult to Annotate

The studies selected under these criteria describe a body of text whose linguistic properties shape every later stage of the review, and those properties are therefore examined first. Three of them affect the annotation task directly. The first is that switching occurs inside the sentence and offers little reliable cue to where the opinion is located. The second is that affective meaning is often carried by function words that appear in no standard sentiment lexicon. The third is that phonetic spelling multiplies the surface forms of a single word. All three are already described in the sociolinguistic literature on Malay-English contact (Treffers-Daller et al., 2022; AlAfnan et al., 2025; Muysken, 2013). What this section adds is not the description but the consequence that each property carries for the annotator, whether human or automatic, who must assign a sentiment label. Each property is therefore presented together with a short statement of that consequence, and the linguistic description itself is not extended beyond what the consequence requires.

2.1 Switching occurs within the sentence and offers little reliable structural cue

Malay-English switching occurs predominantly within the clause rather than between clauses. Treffers-Daller et al. (2022) analysed 1,044 switched utterances and found that insertion, in which a content word or phrase from one language is embedded in a stretch of the other, accounted for 33.0% of switches. Congruent lexicalisation, in which the grammar and vocabulary of both languages operate within a shared word order, accounted for a further 30.7%. Clause-level alternation accounted for 24.8%. The authors explain this distribution typologically rather than stylistically. English and Malay share Subject-Verb-Object (SVO) word order, and Malay carries very little inflectional morphology, so a speaker can move between the two languages mid-clause with minimal grammatical adjustment (Treffers-Daller et al., 2022; Muysken, 2013).

The corpus analysed is spoken, so the exact proportions should not be transferred to social media text. The structural explanation, however, is a property of the language pair rather than of the medium, and it therefore applies wherever the two languages are used together. A written instance is the post "Does institut kanser offer free medication cheap chemotherapy", in which a Malay noun phrase, in Malay head-initial word order, is embedded within an English question (Mohd Shamsuddin et al., 2024).

The consequence for annotation is that the annotator cannot use a language boundary to locate the part of the message that carries the sentiment. In monolingual

annotation this location step is separate from the judgement itself and largely mechanical. Annotation schemes for aspect-based sentiment analysis, for example, require the opinion target expression to be marked as an explicit span in the text before a polarity label is assigned to it (Pontiki et al., 2016; Nazir et al., 2022). In Malay-English mixed text, identifying that span is already an act of interpretation. Two annotators may therefore assign different labels not because they read the sentiment differently, but because they are reading different parts of the message. Disagreement about which span to judge precedes, and is separate from, disagreement about polarity.

2.2 Affective meaning is carried by particles that belong to neither standard language

Locating the opinion-bearing span is only the first difficulty. A second arises once the span has been found. In Malay-English mixed writing, the element that carries the affect is often a Malay particle attached to a clause that is otherwise entirely in English (Treffers-Daller et al., 2022). The construction is a distinct type of code-switching, which Muysken (2013) terms back-flagging. Treffers-Daller et al. (2022) define it as the attachment of a discourse marker or co-ordinate conjunction from a speaker's first language to a stretch of speech in the second. Back-flagging accounts for 121 of the 1,044 switched utterances in their data, or 11.6%. The particle involved most frequently is *lah*, which the same study reports can attach to a clause that is entirely English, entirely Malay or mixed. Two examples it reproduces from earlier descriptive work are "Buy this lah" and "you understand lah kan".

What these particles contribute is pragmatic rather than propositional, and the pragmatic work they do is affective. AlAfnan et al. (2025) classify particles as a lexical feature that adds emphasis or tone and signals politeness or informality. The same study identifies the softening of directives as one of their pragmatic functions, giving "Don't disturb la, he's busy" as an instance in which the particle makes an instruction less direct. Treffers-Daller et al. (2022) describe the effect more precisely: *lah* shifts an utterance from a highly assertive tone to a polite request or a friendly encouragement. These particles belong to Manglish rather than to either source language, and AlAfnan et al. (2025) list them among the features that distinguish Manglish from standard English and standard Malay. In both accounts the particle leaves unchanged what the sentence states. What it changes is how the sentence is to be taken, which is the property a sentiment label is meant to record.

Two limits on this evidence should be noted, and they run in opposite directions. Treffers-Daller et al. (2022) supply frequencies, but from classroom speech. Their corpus is fifteen hours of recorded teaching by each of two university English lecturers, so the rates should not be read as rates for social media writing. AlAfnan et al. (2025) describe Malaysian usage more broadly, but their data are scripted dialogue from Malaysian television series such as *Kopitiam* and *KL Gangster*, examined through qualitative content analysis. That study therefore reports what the particles do and not how often they occur. Neither source establishes a frequency for particle use in Malaysian social media posts, and no such figure is currently

available. What both sources do establish is that the construction is regular and that it carries affect.

The consequence for annotation is that the particle fixes the tone of a message without fixing its polarity. A reader of "Don't disturb la, he's busy" can see that the particle softens the instruction. Softening is nonetheless compatible with a negative reading, in which the writer is irritated, and with a neutral one, in which the writer is being polite. Treffers-Daller et al. (2022) concede the same point about the category as a whole, observing that the specific meanings of discourse markers are often difficult to capture even where their pragmatic functions can be evaluated. The affective signal is therefore carried by a token whose contribution is real but underdetermined, and two annotators may read it differently while both reading it competently.

Disagreement of this type is not removed by adding annotators. Pavlick and Kwiatkowski (2019) show that where a task admits more than one defensible reading, disagreement between annotators is reproducible and persists as further ratings are collected, rather than resolving toward a single label. A corpus built by majority vote over such items therefore records the majority reading as though it were the only defensible one. The variation it conceals is systematic rather than random. An agreement coefficient computed over these items registers genuine interpretive difference and not annotator error.

The first property examined in this section produces disagreement about which part of a message to judge. The second produces disagreement about how to judge a part whose reading is genuinely open.

2.3 Phonetic spelling multiplies the surface forms of a single word

The first two properties concern where the sentiment is and how it is to be read. The third concerns whether the words carrying it can be recognised at all. Malaysian users spell phonetically rather than to a dictionary standard, so a single word appears in many written forms. Maskat and Rahman (2020) provide a formal categorisation of this variation for Malay. Working from a corpus of 6,241 Twitter posts collected between 2014 and 2018, they sort what they call Social Media Malay Language into five categories. The five are spelling variants, Malay-English mixed sentences, loan words written with Malay phonology, slang-based spellings, and vowel-less words.

The variation within each category is wide. Among the spelling variants Maskat and Rahman (2020) record, *macam mana* appears as *mcmana*, *camne*, *camner*, *mcmner* and *cane*. The vowel-less category reduces a word to consonants or to a single character, so *jangan* is written *jgn* and *pada* is written *pd*. The slang category adds a regional layer, with *besar* written *besaq* in the north and *beso* on the east coast. The categories also combine, and the same study gives *x suko* as a form drawing on the slang and vowel-less categories at once.

Whether this variation can be undone automatically is the question that matters for annotation, and Maskat and Rahman (2020) tested it directly. They attempted to normalise two of the five categories, spelling variants and vowel-less words, using n-gram similarity indices and Levenshtein distance. Their benchmark was a ground truth of 1,176 similar-word pairs covering 435 unique words, identified by hand by

a native Malay speaker. Measured against that benchmark, the highest precision reported for any configuration is 0.43, and most configurations fall well below it. The authors conclude that similarity matching can detect the two categories but that a more sophisticated technique is needed, and they state that normalisation of the remaining three categories is open for future research.

Two limits on this evidence should be noted. Only Selangor slang was handled, so the slang category is represented more narrowly than the regional range the same paper describes. More consequentially for this review, English words were removed before the experiment and code-mixed posts were excluded from the evaluation subsets. The reported precision is therefore a measurement on Malay text with the English stripped out, not on Malay-English mixed text, and no equivalent measurement for the mixed case is available.

The consequence for annotation is an asymmetry between the two types of annotator, and it is visible inside the study just described. A single native speaker recovered 1,176 word pairs by hand, reliably enough for the result to serve as ground truth. Automated matching against that same ground truth reached a highest precision of 0.43. A human reader recovers *x* as *tidak* without effort. The automatic procedure largely does not. The two types of annotator therefore do not fail on the same inputs, and they do not fail at the same rate.

That asymmetry changes what the choice of annotator is. If human and automatic annotation differed only in cost and speed, the choice would be a matter of budget. Because they fail on different inputs, the choice determines which errors a corpus will contain, and a corpus labelled entirely by one type carries that type's systematic weakness throughout.

The three properties examined in this section therefore place three distinct demands on annotation. The first makes the opinion-bearing span difficult to locate. The second leaves its reading underdetermined once located. The third makes the tokens themselves unrecoverable to an automatic annotator while they remain transparent to a human one. Section 3 examines the annotation strategies used in response, and the trade-offs each one accepts.

3. Annotation Practice and Its Failure Modes

Section 2 described what makes Malaysian code-mixed text hard to label. This section examines what has been done about it. The strategies reported in the reviewed studies are set out first, arranged by the trade-off each one accepts. The distribution of labels they produce is examined next, and the practice of pooling several annotators is examined last, because it is the remedy most often proposed, and the one the regional evidence tests most directly (Artstein and Poesio, 2008; Uma et al., 2021).

3.1 Annotation strategies and the fidelity-scalability trade-off

For low-resource code-mixed text, label fidelity and scale are in tension. A corpus can be labelled carefully or it can be labelled in volume. The strategies reported by the studies reviewed here are best read not as competing solutions to one problem, but as different decisions about which of the two to sacrifice. On that reading, the

strategies fall into three tiers: manual annotation, semi-automated annotation, and fully automated generative annotation.

The first tier is manual annotation by people. Two of the four regional corpora were built this way, and they are the only two whose labels come from human judgement. Kasmuri and Basiron (2019) drew 5,849 sentences from 23 personal blogs, selected 4,191 for annotation, and retained 2,316 in the finished corpus, reporting $\kappa = 0.83$ for the judgement of whether a sentence was opinionated or factual. Kong et al. (2023) collected 108,246 tweets and manually annotated 11,568 of them into three sentiment classes, using three independent annotators working from a written guideline. What both illustrate is the cost of the decision rather than a defect in the method. Kasmuri and Basiron kept 40% of the sentences they started with, and Kong et al. labelled 11% of what they collected. The labour that produces reliable labels is the same labour that prevents either corpus from reaching the scale a general-purpose Manglish resource would require.

The second tier is semi-automated annotation, which trades in the opposite direction by using existing tools to label at volume and accepting the tools' limitations as the price. Mohd Shamsuddin et al. (2024) built MESocSentiment this way. They collected 229,566 tweets carrying the "#Malaysia" keyword, reduced the set to 138,646 in pre-processing, applied the Malaya and Lingua libraries to identify code-switched posts, and arrived at 20,226 tweets. Sentiment labels were then assigned by majority vote among three existing analysers. The authors state the motivation plainly. The language identification libraries "reduced the need to find fluent speakers in Malay and English to classify the tweets". The reasoning is defensible, and there is evidence from other labelling tasks that automated annotators can match or exceed human crowd workers (Gilardi et al., 2023). What the tier concedes is that the labels can be no more competent than the tools producing them, and inherit whatever those tools cannot read.

The third tier replaces those tools with generative models used directly as annotators. Because a generative model is not restricted to a fixed vocabulary, it can in principle interpret an unfamiliar spelling or an unseen slang term from context, which is the capability the second tier lacks. In principle this dissolves the trade-off. The peer-reviewed evidence does not yet support that reading. Jadhav et al. (2025) evaluated generative annotators on Marathi, an Indo-Aryan language of western India, across four tasks: sentiment analysis, two-class hate speech detection, four-class hate speech detection, and news classification. They found two generative large language models, GPT-4o and Llama 3.1 405B, trailing a fine-tuned BERT baseline by average accuracy margins of 10.2% and 14.1% respectively. BERT is a bidirectional transformer encoder pre-trained on unlabelled text. They conclude that fine-tuned transformers remain necessary where linguistic resources are scarce. Marathi is morphologically rich where Malay is not, so the result bounds the tier's promise rather than measuring it in this setting. That it is the only peer-reviewed study of the question available to this review is itself part of the finding. The third tier is the least evidenced of the three, and no study examined here evaluates generative annotation on Malaysian code-mixed text.

Table 1. Annotation Strategies for Malaysian Malay-English Code-Mixed Sentiment Data

Tier	Regional example	Scale achieved	Evidence of fidelity	What is conceded
Manual	Kasmuri and Basiron (2019)	2,316 sentences retained from 5,849 candidates, drawn from 23 blogs	$\kappa = 0.83$, for a binary opinionated-or-factual judgement	Scale
Manual	Kong et al. (2023)	11,568 tweets annotated from 108,246 collected	Three annotators and a written guideline; no coefficient reported	Scale
Semi-automated	Mohd Shamsuddin et al. (2024)	19,714 posts from 229,566 collected	None reported; the pipeline has no human annotation stage	Demonstrable fidelity
Fully automated, generative	None published for Malay-English	—	—	Reliability precisely where linguistic resources are scarce (Jadhav et al., 2025)

3.2 Two causes of neutral-label skew

The clearest regional evidence of what the second tier costs is the class distribution of the corpus it produced. MESocSentiment is labelled 78.23% neutral, 16.33% positive and 5.44% negative across 19,714 posts (Mohd Shamsuddin et al., 2024). Two reasons support reading that distribution as an artefact of the labelling method rather than a property of Malaysian discourse. The corpus was collected under the "#Malaysia" keyword across sports, politics and the arts, which are not topics on which social media users are typically neutral. A comparable corpus of Malaysian tweets labelled by human annotators is only 23.6% neutral (Kong et al., 2023). This section separates the two distinct causes at work in that skew, because they produce identical labels and require opposite responses.

The first cause is the behaviour of a scoring function that finds nothing to score. Lexicon-based sentiment tools work by matching each word or phrase in a text against a sentiment lexicon and aggregating the scores of the matched words into an overall score (Selvaraju et al., 2024). That score is then divided into classes by threshold. VADER, one of the three analysers used to build MESocSentiment, sets its classification thresholds at -0.05 and $+0.05$ on a normalised score running from -1 to 1 (Hutto and Gilbert, 2014). Two consequences follow from combining those two facts, and this review states them as an inference rather than as a reported finding. A text whose affective tokens are absent from the lexicon contributes little or nothing to the aggregate. Its score therefore remains within the neutral band, and it is returned as neutral. On that reading, neutral is not always a judgement that a text is affectively balanced. It can also be the default output when nothing was recognised.

The corpus authors' own error analysis indicates that the mechanism operates on this data. Alongside the corpus, Mohd Shamsuddin et al. (2024) present two small tables comparing the ensemble's label with the correct one. All three of the misclassified examples they give are neutral labels that should have been positive, and in all three the sentiment is carried by Malay: *memang cantik sangat*, *selamat deepavali* and *raih gelar juara*. In the correctly classified negative example, the sentiment is carried by an English word, *worthless*. Six author-selected examples are illustrative rather than statistical, and no error rate can be computed from them. Their uniformity is nonetheless difficult to dismiss: where the ensemble succeeded, the affect was in a language its tools covered, and in every case where it failed, the affect was not.

The second cause has nothing to do with tooling. Some posts are genuinely hard to judge, and the annotators of the regional corpora say so. Kasmuri and Basiron (2019) note that low agreement in sentiment annotation arises because sentiment texts are subjective, so evaluations differ from one annotator to another and are shaped by each annotator's background. Kong et al. (2023) make the same point about their own task, observing that tagging text with sentiment labels is highly subjective and influenced by personal beliefs, and that a clear guideline is needed for this reason. As Section 2.2 established, disagreement of this kind is not removed by collecting more judgements (Pavlick and Kwiatkowski, 2019).

Separating the two causes matters because the labels they produce are indistinguishable and the responses they require are opposite. A genuinely neutral post is correctly labelled and belongs in the corpus. A falsely neutral post is an error that a better-resourced annotator would not have made. A single-model pipeline is structurally unable to tell them apart, because the evidence that would distinguish them, namely the affect the annotator could not read, is exactly the evidence that annotator does not have.

The consequence is more damaging than the aggregate percentage suggests, because a false neutral does not only inflate one class. It also depletes another. Every strongly polarised post absorbed into the neutral class is a training example removed from a minority class that a downstream classifier needs most. At a 5.44% negative share, the smallest class in the corpus is also the one most exposed to that erosion.

3.3 What consensus does with disagreement

Pooling several annotators is the response most often proposed when a single annotator is unreliable (Artstein and Poesio, 2008; Uma et al., 2021). MESocSentiment is evidence against that response, because it already was a consensus pipeline. Three sentiment analysers labelled every post and the majority label was assigned. The corpus skewed anyway.

The reason is visible in the corpus authors' own reporting, and it is not the one that might be expected. Voting cancels error only to the extent that voters fail independently, and the usual guarantee of independence is architectural difference. Table 2 shows that architectural difference was present and did not help.

Table 2. The Three Annotators Used to Build MESocSentiment, and the Labels Each Assigned (Mohd Shamsuddin et al., 2024)

Analyser	How the corpus authors describe it	Languages it is described as covering	Neutral labels assigned, of 19,714
TextBlob	"a Python library used to process textual data"	Not stated	14,305 (72.6%)
VADER	"a rule-based model … made to focus on social media texts"	Not stated	13,115 (66.5%)
pysentimiento	A toolkit "trained for sentiment analysis in English, Spanish, Portugeuse, and Italian"	English, Spanish, Portuguese, Italian	16,916 (85.8%)
Majority vote	—	—	15,422 (78.2%)

Two things follow. The first is that every analyser skewed neutral before any vote was taken, so the skew was not produced by the voting rule. The second is that the analyser skewing hardest, at 85.8%, is pysentimiento, which is the transformer-based tool of the three, built on the neural architecture that underlies most current language models. The shared weakness is therefore not shallow processing, since the deepest model performed worst. None of the three is described by the corpus authors as covering Malay, and one is explicitly described as trained on four other languages. The property the three voters share is the language they cannot read, and a Malay affective token invisible to one is invisible to all. The ensemble does not cancel that error. It ratifies it, and unanimity produced this way is indistinguishable from confidence.

The second weakness concerns the posts the ensemble cannot agree on. Where all three analysers returned different labels there was no majority, and the authors state their response directly: ambiguous cases "were removed to ensure the corpus only contains tweets from absolute categories". This affected 512 posts, or 2.53% of the 20,226 labelled. The difficulty is not the proportion but the composition. A post on which three analysers disagree is one where the signal is contested, or is carried by material none of them reads confidently. That description fits the heavily code-switched and particle-laden cases Section 2 identified as the hardest to annotate. The rule therefore removes the examples a robust classifier would most need, and removes them systematically rather than at random.

The effect on the corpus can be computed from the paper's own tables, and it runs in the wrong direction. Removing the contested posts left the neutral count unchanged at 15,422 while reducing the total from 20,226 to 19,714, so the neutral share rose by roughly two percentage points. Taken with the first weakness, static voting has a double effect: it keeps the labels it is confident about and wrong about, and deletes the ones it is uncertain about and would have learned from.

The methodological literature on annotation supports treating that deletion as a loss rather than a cleanup. Uma et al. (2021) question the assumption that a single correct interpretation exists for every item. On that view, the 512 discarded posts were not noise removed from the corpus but signal removed from it.

What the reviewed literature does not contain is any evaluation of a better procedure on this data. No study examined here adjudicates annotator disagreement on Malaysian code-mixed sentiment text, and the premise that annotators of different types fail independently, which is what licenses aggregation in the first place, is asserted rather than measured on this material. The evidence assembled above suggests that independence fails when the annotators share a language gap, but a suggestion is not a measurement and this review does not treat it as one. Both points are recorded here and returned to in Section 6 as questions for future research.

4. Models as Annotation Instruments

Section 3 established that the choice of annotator determines which errors a corpus will contain. This section asks where each family of model is limited when used as an annotator. Two families are examined: fine-tuned multilingual encoders, and generative models used directly as annotators. The question is not which architecture classifies best, which belongs to the comparative benchmarking placed outside the scope in Section 1. It is what each family cannot read, because that limitation is inherited by every corpus it labels.

Fine-tuned multilingual encoders acquire their competence from exposure. An encoder here is a model that reads text and produces representations of it, rather than generating new text. Manukonda and Kodali (2025) illustrate the mechanism. They fine-tuned XLM-RoBERTa, a multilingual encoder model, over original, fully transliterated and partially transliterated versions of the same text, and report a macro F1 of 0.90 on the shared-task test set. The training method was masked language modelling, in which the model predicts words hidden from the input. Macro F1 is the mean of the per-class F1 scores, and F1 is the harmonic mean of precision and recall. The model handled the variant spellings because it had been shown them in training. A corollary follows from the method rather than from any claim the authors make. An encoder resolves the variation it has been trained on, so the coverage of a fine-tuned annotator is bounded by the coverage of its fine-tuning data. Extending that coverage requires more annotated or transliterated regional data, which is the resource whose scarcity motivated automated annotation in the first place. Two limits should be noted: the study is a shared-task system description rather than a full investigation, and the task is fake news detection in Malayalam rather than sentiment analysis in Malay. What transfers is the mechanism, not the result.

Generative models locate their competence differently. Section 3.1 described the capability that follows from not being restricted to a fixed vocabulary. It is the capability the encoder lacks, and it is why generative annotation is attractive for a variety that changes faster than its resources. The competence is distributed unevenly across languages, and not in the way resource volume alone would predict. Jadhav et al. (2025) found frontier models trailing a fine-tuned BERT baseline by 10.2% and 14.1% accuracy when annotating Marathi. Adilazuarda et al. (2022) tested Indonesian code-mixed with four embedded languages, namely English, Sundanese, Javanese and Malay, and found robustness to English mixing lower than robustness to Javanese and Malay mixing across all models tested. They attribute the difference to the composition of the pre-training corpus. The direction of that

result should be stated explicitly. The embedded language best represented in pre-training was the one the models handled least robustly. Robustness to code-mixing therefore does not follow directly from how much of a language a model has seen.

The operative point is not the relative standing of the two families but the shape of their respective limitations. The encoder is reliable on what it has been shown and blind to what it has not. The generative model reasons about the unfamiliar, but its robustness on a given language pair cannot be predicted from the volume of data behind either language. Selecting an annotator is therefore not selecting a level of quality. It is selecting a failure mode, and the labels a corpus contains carry the signature of whichever mode was selected.

5. Evaluating Annotation Reliability

Sections 3 and 4 concerned how labels are produced. This section concerns how the labels that result are shown to be sound. Reliability is normally demonstrated with a chance-corrected agreement coefficient, of which Cohen's κ is the most widely used (Cohen, 1960; Artstein and Poesio, 2008). The coefficient measures how far the agreement two annotators actually reach exceeds the agreement expected by chance. Magnitudes are then read against conventions. In computational linguistics, values above 0.8 are generally treated as good reliability, and values between 0.67 and 0.8 as allowing tentative conclusions (Artstein and Poesio, 2008).

Those conventions assume that expected chance agreement is stable across datasets, and it is not. Feinstein and Cicchetti (1990) show that an imbalance in the marginal totals of an agreement table can convert a high observed agreement into a low coefficient. Their worked example illustrates the effect. Holding observed agreement at 0.85, an expected chance agreement of 0.50 yields $\kappa = 0.70$, while an expected chance agreement of 0.78 yields $\kappa = 0.32$. The annotators agreed on the same 85% of items in both cases. Byrt et al. (1993) formalise the effect, showing that κ reflects the distribution of data across categories as well as agreement, and propose a prevalence-adjusted coefficient in response. Artstein and Poesio (2008) state the consequence for annotation directly: when one category is very common, high agreement can result from indiscriminate coding rather than from an ability to distinguish between categories.

That is the condition Malaysian code-mixed sentiment data is in. Section 3 established that the labelling methods available for this text produce heavily skewed class distributions, the largest corpus standing at 78.23% neutral. A skew of that size raises expected chance agreement, which is the quantity the correction depends on. The property that makes the labels unrepresentative is therefore the property that makes the statistic used to check them difficult to read. The difficulty runs in both directions. A low coefficient may indicate unreliable annotation, or a dominant class. A high one may indicate reliable annotation, or indiscriminate coding. Neither reading can be ruled out from the coefficient alone. This is the sense in which the annotation problem and the evaluation problem are one problem rather than two, and the reason correcting either alone leaves the other in place.

What the four regional corpora report about their own reliability is set out in Table 3.

Table 3. Reliability Evidence Reported by the Four Malay-English Sentiment Corpora

Corpus	Annotators	Written guideline	Coefficient reported	What the coefficient measured
MESocSentiment (Mohd Shamsuddin et al., 2024)	None; three automatic analysers	Not applicable	None	—
MyCovid-Senti (Kong et al., 2023)	Three, independent	Yes	None	—
MY-EN-CS (Kasmuri and Basiron, 2019)	Two undergraduates proficient in Malay and English	Yes; the guideline is the paper's contribution	$\kappa = 0.83$	Whether a sentence is opinionated or factual
SentiBahasaRojak (Romadhona et al., 2022)	Five native Bahasa Rojak experts	Not stated	"Kappa" = 0.77; the variant is not specified	Whether a stock-market post is positive or negative; reported for the stock review subset only

Three observations follow. Half of the regional corpora report no agreement coefficient at all, including one annotated by three people working from a written guideline. The two coefficients that are reported were obtained for binary judgements, and neither is the three-class polarity task the corpora are used to support. Romadhona et al. (2022) report a kappa value without stating which coefficient it is. Their annotation used five experts with the final label set by majority vote, and Cohen's κ is defined for two annotators.

None of the four corpora examined here therefore supplies what a reliability claim requires: a coefficient of a stated type, computed for the task the labels are used for, over annotators whose number and qualification are recorded. A corpus that did supply them would still face the interpretation problem above, because the class distributions these methods produce are the distributions under which the coefficient becomes hard to read. Establishing reliability for this data is not only unreported. With the instruments currently in use, it is not clearly achievable.

6. Synthesis: How the Challenges Compound

Sections 2 to 5 examined linguistic structure, annotation practice, model capability and reliability measurement in turn, which is how the studies reviewed here are organised. Read separately they describe four difficulties. Read together they describe one sequence, in which each difficulty creates the conditions for the next and the sequence returns to its own starting point. Figure 1 sets it out.

Establishing that this structure exists is the contribution this review claims, and the directions in the second half of this section are traced to points within it.

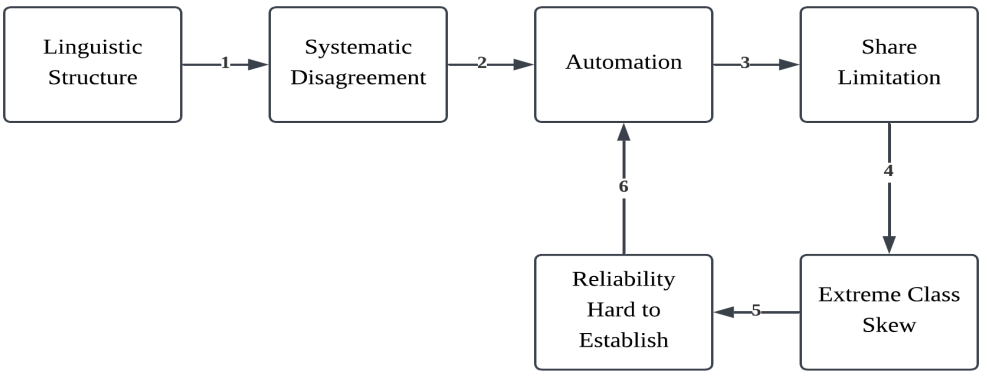


Figure 1. The Compounding Chain in Malaysian Code-Mixed Sentiment Annotation

The first link runs from linguistic structure to systematic disagreement. Section 2 established three properties of this text. The consequence is not only that agreement is harder to reach. It is that disagreement takes a particular form. Where one annotator reads *lah* as a softener and another reads it as a marker of sarcasm, and each reading is stable, the disagreement recurs on the same construction in the same direction. Random disagreement is reduced by recruiting more annotators, because independent errors average out. Systematic disagreement is not, because additional annotators bring their own stable readings rather than noise around a shared one (Pavlick and Kwiatkowski, 2019). The usual response to unreliable annotation is therefore unsuited to the type of unreliability this text produces.

The second link runs from systematic disagreement to automation. Competent annotation of this material requires a fluent bilingual reader, and that requirement follows from the linguistic properties of the text rather than from any general fact about annotation. The corpus papers give cost and annotator availability as their reason for automating, and Mohd Shamsuddin et al. (2024) state that their tools "eliminated the need to find speakers fluent in Malay and English to label the dataset". Automation on this material is therefore downstream of the linguistic difficulty rather than an independent methodological preference.

The third link runs from automation to a shared limitation. The tools available for automation are those built for well-resourced languages. Section 3.3 documented the one regional case: none of the three analysers covers Malay, each skewed neutral before any vote was taken, and voting cancels error only to the extent that voters fail independently. Where the property they share is the language none of them reads, the ensemble ratifies the error instead of cancelling it.

The fourth link runs from the shared limitation to extreme class skew. False neutrals accumulate rather than being corrected, and the corpus acquires a class distribution that no participant intended. MESocSentiment reaches 78.23% neutral. Posts on which the three analysers could not agree were removed, 512 of them, on the stated ground of retaining only "absolute categories". Those posts are not a

random sample. They are the contested, heavily code-switched cases that the first link identifies as the source of the difficulty, and removing them raised both the neutral share and the measured agreement.

The fifth link runs from extreme class skew to the difficulty of establishing reliability, and it is where the two halves of this review meet. The distribution produced at the fourth link is the specific condition under which a chance- corrected coefficient stops returning interpretable values (Feinstein and Cicchetti, 1990; Byrt et al., 1993). A researcher who suspects the labels are unsound and computes a coefficient obtains a low value, which is consistent with poor annotation and equally consistent with competent annotation of a skewed corpus. Raw agreement carries the same ambiguity in the opposite direction (Artstein and Poesio, 2008). The defect and the inability to detect the defect have one cause. That is what separates an unresolved problem from a merely difficult one: here the quality check is least informative exactly where quality is lowest.

The sixth link closes the cycle. Corpora whose reliability was never established are released, trained on and evaluated against, so every downstream comparison inherits an error term that nobody has quantified. The more consequential edge is the one that returns. The annotators used to build the next corpus are fine-tuned on the last, and Section 4 showed that a fine-tuned encoder resolves only the variation it has been shown. The limitations of one generation of labels are transmitted to the instruments that produce the next, so the chain does not merely persist. It reproduces itself.

Read as a whole, the chain explains a pattern that the individual studies cannot. Each reviewed work intervenes at one link and does so competently. None addresses the chain as a chain, and an intervention at one link is assessed with an instrument compromised at another, so the assessment cannot establish whether the intervention worked. That is why effort in this area has been absorbed without accumulating.

The chain identifies where intervention would matter, not which intervention would succeed. Each of the five directions below is traced to a link, since a direction that cannot be traced to a finding is a preference rather than a conclusion.

The first direction is agreement measurement under class skew, and it follows from the fifth link. Prevalence-adjusted coefficients exist (Byrt et al., 1993) but have not been evaluated on code-mixed sentiment data, so it is not known which behaves best under the distributions these corpora actually exhibit. The study requires no new data collection, only an existing annotated corpus, a set of candidate coefficients, and a comparison of how they behave as skew increases. It is a precondition for the others, because until reliability can be measured under prevailing conditions, no corpus built hereafter can be shown to improve on the ones described here.

The second direction is a minimum reporting standard, and it follows from Table 3. A minimum set for any newly released corpus would comprise five elements: (1) the number of annotators and their stated qualification; (2) the availability of the annotation guideline; (3) the variant of the coefficient and the task it was computed for; (4) per-class rather than aggregate agreement; and (5) the full class distribution.

Per-class agreement deserves particular emphasis, since an aggregate figure on a skewed corpus is dominated by the majority class and can conceal near-total disagreement on the minority polarities that matter most.

The third direction is a shared annotation convention for affect-bearing particles, and it follows from the first link. Because the disagreement these particles generate is systematic rather than random, the appropriate remedy is a shared convention rather than additional annotators, and no such convention exists for *lah*, *kan* and their equivalents. A model is available. Mohammad (2016) proposes a questionnaire that separates the speaker's emotional state, the target of the opinion and the attitude taken toward it. It also treats sarcasm, ridicule and mockery as a distinct response rather than folding them into polarity, which is precisely the ambiguity Section 2.2 identified. Kong et al. (2023) adapt this scheme into a four-question guideline for Malaysian tweets. What is absent is not the design but its adaptation to Malaysian particles and its adoption across projects.

The fourth direction is to publish label distributions rather than only resolved labels, and it follows from the fourth link. Current practice resolves disagreement by majority rule or removes it, discarding the information that the case was contested. Uma et al. (2021) report that models trained on the full distribution of annotator judgements can outperform models trained on aggregated labels. Publishing that distribution would preserve the cases that current pipelines delete. It would also make the fourth link visible, since a corpus that reports what it discarded allows a reader to judge whether its reported reliability was earned or obtained by attrition.

The fifth direction is to test the independence premise, and it follows from the third link. Every consensus mechanism in this literature depends on the premise that annotators of different architecture fail independently, and on this material the premise is asserted throughout and measured nowhere. The study is a direct one: label a common sample with annotators of deliberately different competence, and measure the correlation of their errors rather than their individual accuracies. Low correlation would justify the aggregation strategies already in use. High correlation would indicate that consensus on this material adds confidence without adding evidence.

These are directions rather than results. The claim of this review is not that they will succeed, but that Figure 1 identifies where effort in this area is currently absorbed without accumulating.

7. Conclusion

This review examined one question: why reliable annotation and evaluation of Malaysian Malay-English code- mixed sentiment data remain unresolved. The answer usually given is architectural, and the evidence assembled here indicates that it is incomplete. Model choice accounts for part of the difficulty. The rest originates earlier, in the sentiment labels themselves and in the evidence offered that those labels are sound.

The contribution this review claims is not any of those findings taken alone but the relation between them. The linguistic properties of the text produce

disagreement that is systematic rather than random, which makes fluent bilingual annotation necessary and expensive, which is why labelling is automated. The tools available for that automation cover neither Malay nor Manglish, so pooling them ratifies a shared error instead of cancelling it. The class skew that results is the precise condition under which the standard agreement coefficient stops returning interpretable values, so the defect and the inability to detect it have one cause rather than two. Each corpus built this way then trains the annotators that build the next. An intervention at any single point is assessed by an instrument compromised at another, which is why competent work at each link has not accumulated into progress.

The directions set out in Section 6 follow from that structure rather than from preference. The first of them, an agreement measure that behaves interpretably under the class distributions these corpora actually exhibit, is a precondition for evaluating the rest. Until reliability can be measured under the conditions this data presents, no corpus built hereafter can be shown to improve on the ones examined here.

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